

Practicum 1

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1 Exercise 1– Compose program

Compose a program that solves the steady convection-diffusion equation

$$U \frac{d\phi}{dx} = k \frac{d^2\phi}{dx^2} \quad (1)$$

on $0 \leq x \leq L$, with

$$\phi(0) = \phi_0, \quad \phi(L) = \phi_1. \quad (2)$$

Take into account the following:

1. As input variables, the program should use values of U , k , ϕ_0 , ϕ_1 and the number of segments N covering the domain $[0, L]$.
2. Various discretisation methods should be possible, e.g. central, (smart-)upwind.
3. Very fine grids should be possible, e.g.

$$h = \frac{L}{N}, \quad (3)$$

with $N = 100,000$.

Hence, you cannot store the full 2D coefficient matrix A and you cannot use linear solver commands (in Matlab) such as `inv(A)*b` or `A\b`. Instead, you should only store the 3 diagonals of A and use the TDMA algorithm as provided (in Matlab), or use `scipy.sparse.diags` (in Python).

Remark: during other practica we will include more discretisation options (e.g. 2nd order upwind, Quick), non-equidistantly spaced grids and the unsteady term $\frac{d\phi}{dt}$.

1.1 Implementation Approach

Before applying the solver to the exercises, several verification cases are performed to ensure that the implementation is correct. These tests validate the assembly of the discretized equations, the boundary-condition treatment, and the TDMA solution procedure. In addition, a modular code structure is adopted to facilitate the incorporation of alternative discretisation schemes in later exercises. The overall workflow of the program is summarized in the following pseudocode.

1.1.1 Solver Pseudocode

1. Read input parameters:

$$U, k, \phi_0, \phi_1, N.$$

2. Generate a one-dimensional mesh.
3. Select the discretisation scheme.

4. Assemble the coefficient arrays

$$a_W, a_P, a_E,$$

and the right-hand side vector.

5. Apply boundary conditions.
6. Construct the three TDMA diagonals.
7. Solve the linear system using TDMA.
8. Reconstruct the full solution vector.
9. Post-process and visualize the results.

1.2 Code Verification

Before performing the studies requested in the assignment, the implementation was verified for all discretisation schemes available in the solver. The objective of this verification step is to ensure that the coefficient assembly, boundary-condition treatment, and TDMA implementation are functioning correctly for each discretisation method.

For all verification cases, the same physical problem was considered:

$$L = 10, \quad U = 50, \quad k = 10^{-3},$$

with Dirichlet boundary conditions

$$\phi(0) = 0, \quad \phi(L) = 20.$$

A highly refined mesh consisting of

$$N = 10^6$$

cells was used for all tests. This ensures that any observed differences are primarily caused by the discretisation scheme rather than insufficient spatial resolution.

1.2.1 Test Case 1: Central Differencing Scheme

The first verification test was performed using the central differencing discretisation.

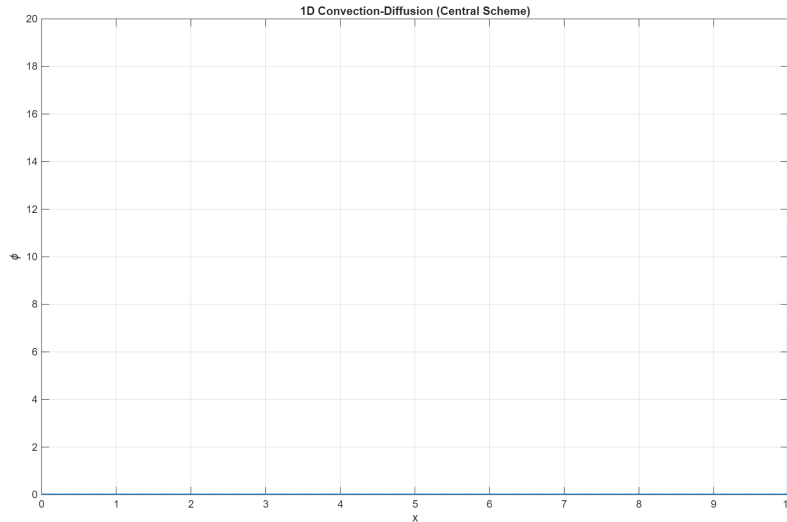


Figure 1: Numerical solution obtained using the central differencing scheme.

The resulting solution exhibits the expected behaviour for a convection-dominated problem and confirms that the coefficient assembly and TDMA implementation are functioning correctly for the central discretisation.

1.2.2 Test Case 2: Upwind Differencing Scheme

The second verification test employed the first-order upwind discretisation.

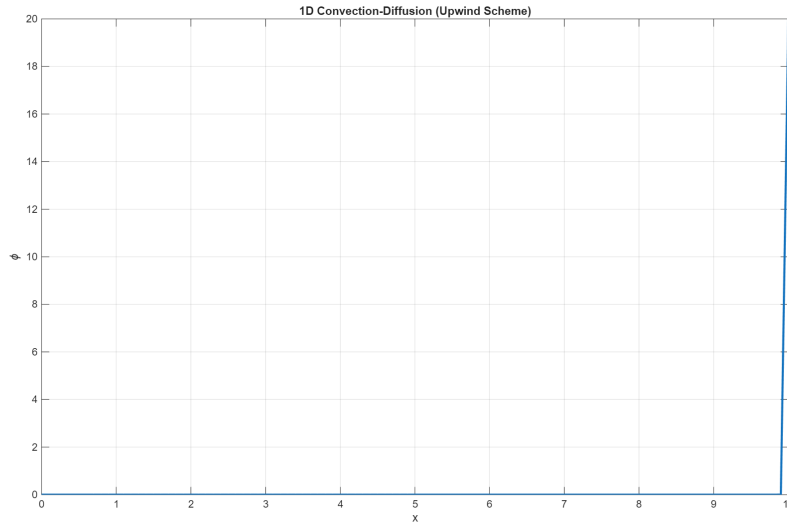


Figure 2: Numerical solution obtained using the upwind differencing scheme.

The computed solution displays the characteristic smoothing associated with numerical diffusion, indicating that the upwind coefficients are correctly assembled and solved.

1.2.3 Test Case 3: Downwind Differencing Scheme

The final verification test was carried out using the downwind discretisation. For the present problem, the convection velocity is positive ($U > 0$), implying transport from west to east. The downwind scheme therefore uses information from the downstream direction, which is opposite to the physical propagation of information.

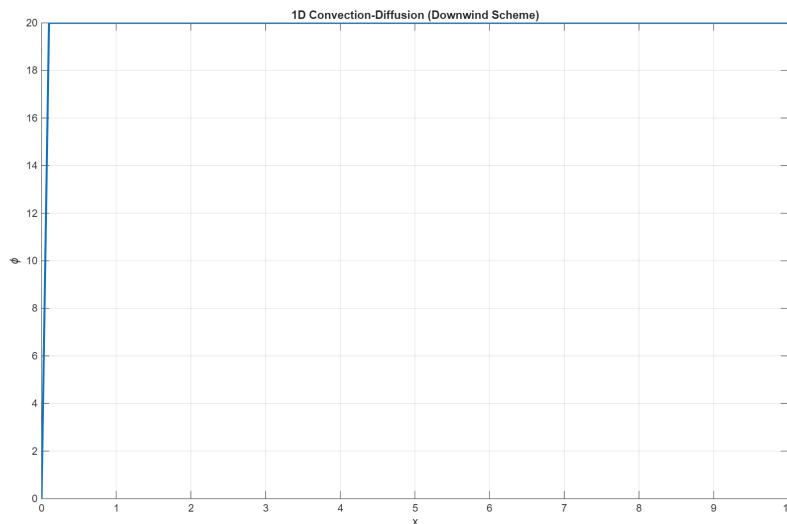


Figure 3: Numerical solution obtained using the downwind differencing scheme.

The resulting solution is highly non-physical. Rather than gradually evolving from the inlet value to the outlet value, the scalar field immediately approaches the outlet boundary value after the first node.

The numerical solution is therefore characterised by a profile of the form

$$\phi \approx [0, 20, 20, \dots, 20].$$

This behaviour is expected because the downwind discretisation propagates information in the wrong spatial direction.

Although the obtained solution is physically incorrect, reproducing this characteristic response provides confidence that the downwind discretisation has been implemented correctly, as well as the convolution-diffusion 1d function, and the corresponding plots.

2 Exercise 2 – Wiggles

Consider the case

$$L = 10, \quad U = 50, \quad k = 10^{-3}, \quad (4)$$

with boundary conditions

$$\phi_0 = 0, \quad \phi_1 = 20. \quad (5)$$

1. When does central discretisation result in a smooth solution without wiggles? Compare your “experimental” result with the theoretical grid-size limit.

To answer this question, two different plots were constructed. The first plot presents the numerical solution for various mesh resolutions, allowing the onset and disappearance of numerical wiggles to be observed directly. The second plot is a log-log convergence plot, where the solutions obtained on different meshes are compared against the analytical solution. The theoretical condition for obtaining a wiggle-free central differencing solution is

$$Pe_h = \frac{U\Delta x}{k} \leq 2,$$

where

$$\Delta x = \frac{L}{N}.$$

Substituting the problem parameters ($U = 50$, $L = 10$, and $k = 10^{-3}$) yields the theoretical grid-size requirement

$$N \geq \frac{UL}{2k} = \frac{50 \times 10}{2 \times 10^{-3}} = 250000.$$

The numerical results are presented in Figure 4, which combines both the solution profiles and the convergence analysis. The left panel shows the numerical solution obtained using central differencing for different mesh resolutions, while the right panel presents the corresponding log-log convergence plot using the analytical solution as the reference solution.

From Figure 4, it can be clearly observed that meshes with $N < 250000$ exhibit the characteristic wiggles associated with central differencing in convection-dominated flows. These wiggles appear as alternating peaks and troughs near the outlet boundary. Moreover, the oscillation amplitude grows exponentially towards the outlet. For coarser meshes, this exponential growth starts earlier in the domain, resulting in larger deviations from the reference solution and consequently larger numerical errors. As the mesh is refined, the onset of the oscillatory behaviour is pushed progressively closer to the outlet boundary and the magnitude of the oscillations decreases.

The theoretical criterion for obtaining a monotone solution with central differencing is

$$Pe_h = \frac{U\Delta x}{k} \leq 2,$$

which, for the present case, corresponds to

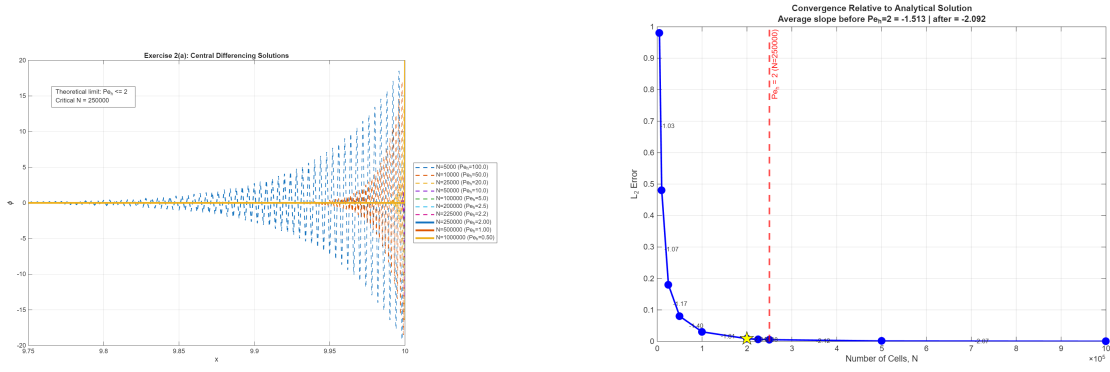


Figure 4: Influence of mesh resolution on central differencing. Left: numerical solution profiles near the outlet boundary. Right: convergence plot relative to the analytical solution.

$$N \geq \frac{UL}{2k} = 250000.$$

The numerical results confirm this prediction. The solution corresponding to $N = 250000$ is the first solution that no longer exhibits visible oscillations, while all coarser meshes display some degree of wiggle behaviour. Further grid refinement produces increasingly smooth solutions and reduces the discretisation error.

The convergence plot in Figure 4 further supports these observations. The average convergence rate before the theoretical limit is noticeably lower than the desired second-order behaviour associated with central differencing. Although the observed slope approaches the theoretical value as the mesh resolution increases, particularly for $N = 200000$, the oscillatory nature of the solution prevents the method from fully exhibiting its asymptotic convergence properties, however the approximate rate is 1.8 which means we are basically there. Once the theoretical limit $Pe_h \leq 2$ is satisfied, the observed slope becomes approximately equal to, or slightly larger than, two, indicating that the scheme has entered its expected second-order convergence regime.

For very fine meshes, the error becomes extremely small and the convergence curve eventually begins to flatten. At this stage, the discretisation error is no longer dominant and the solution enters a region where floating-point round-off errors become comparable to the remaining numerical error. Consequently, further mesh refinement yields only marginal improvements in accuracy.

Therefore, the experimental observations are in excellent agreement with the theoretical grid-size limit. A smooth, wiggle-free solution is obtained when the condition

$$Pe_h \leq 2$$

is satisfied, confirming both the theoretical monotonicity criterion and the expected convergence behaviour of the central differencing scheme.

2. Does upwind discretisation always result in a smooth solution? Find a relation between the obtained boundary layer thickness and the grid size, both experimentally and theoretically, and compare these. When is the solution “accurate enough”? How would you define “enough”?

For $U > 0$, the upwind discretisation uses

$$\left. \frac{d\phi}{dx} \right|_i \approx \frac{\phi_i - \phi_{i-1}}{h}.$$

Substituting this approximation into the convection-diffusion equation and rearranging yields

$$a_W \phi_{i-1} + a_P \phi_i + a_E \phi_{i+1} = 0,$$

with

$$a_W = D + F, \quad a_E = D, \quad a_P = -(a_W + a_E),$$

where

$$D = \frac{k}{h}, \quad F = U.$$

Since $k > 0$ and $U > 0$,

$$D > 0, \quad F > 0,$$

and therefore

$$a_W > 0, \quad a_E > 0, \quad |a_P| = a_W + a_E.$$

Consequently, all off-diagonal coefficients are positive and the resulting discretisation matrix is a monotone operator. Such operators satisfy a discrete maximum principle, meaning that the numerical solution remains bounded between the boundary values and does not exhibit spurious oscillations. Therefore, unlike central differencing, the upwind scheme always produces a smooth, wiggle-free solution for $U > 0$ and $k > 0$. The price paid for this monotonicity is numerical diffusion. Comparing the central and upwind discretisation coefficients reveals the origin of the numerical diffusion introduced by the upwind scheme.

For central differencing,

$$a_W^{CD} = D + \frac{F}{2}, \quad a_E^{CD} = D - \frac{F}{2},$$

where

$$D = \frac{k}{h}, \quad F = U.$$

For upwind differencing,

$$a_W^{UD} = D + F, \quad a_E^{UD} = D.$$

The difference between the two schemes is therefore

$$a_W^{UD} - a_W^{CD} = \frac{F}{2},$$

and

$$a_E^{UD} - a_E^{CD} = \frac{F}{2}.$$

This additional contribution acts as an extra diffusive conductance,

$$\Delta D = \frac{F}{2}.$$

Since the diffusive conductance is defined as

$$D = \frac{k}{h},$$

the equivalent numerical diffusion coefficient k_{num} satisfies

$$\frac{k_{\text{num}}}{h} = \frac{F}{2} = \frac{U}{2}.$$

Multiplying by h gives

$$\boxed{k_{\text{num}} = \frac{Uh}{2}}.$$

Hence, the upwind scheme behaves as if it solves the original equation with an effective diffusion coefficient

$$k_{\text{eff}} = k + k_{\text{num}} = k + \frac{Uh}{2}.$$

The characteristic boundary-layer thickness can be obtained from the analytical solution of the convection-diffusion equation,

$$U \frac{d\phi}{dx} = k \frac{d^2\phi}{dx^2},$$

which has the general form

$$\phi(x) = C_1 + C_2 e^{Ux/k}.$$

The exponential variation occurs over the characteristic length scale

$$\delta = \frac{k}{U},$$

which represents the physical boundary-layer thickness.

Since the upwind scheme introduces the effective diffusion coefficient k_{eff} , it follows that the numerical boundary layer is characterized by

$$\delta = \frac{k_{\text{eff}}}{U}.$$

Substituting the expression for k_{eff} yields

$$\delta = \frac{k + \frac{Uh}{2}}{U} = \frac{k}{U} + \frac{h}{2}.$$

Therefore,

$$\delta = \frac{k}{U} + \frac{h}{2}$$

showing that the boundary-layer thickness increases linearly with the grid spacing. Consequently, coarse meshes produce an artificially thick boundary layer due to numerical diffusion, whereas the physical thickness

$$\delta_{\text{phys}} = \frac{k}{U}$$

is recovered as $h \rightarrow 0$. In other words, the theoretical prediction is a linear dependence of boundary-layer thickness on the grid spacing. For sufficiently coarse meshes, the term $h/2$ dominates and the boundary-layer thickness scales as $\delta \propto h$ meaning experimental measurements should therefore exhibit a slope close to unity when plotted on a log-log scale against the grid spacing, which is precisely what we have in Figure 5.

3. The boundary condition $\phi(10) = 20$ forces the solution towards a fixed value at $x = 10$. The alternative boundary condition (so called Neumann condition) $\phi'(10) = 0$ gives the system more flexibility. This might enhance the occurrence of wiggles, although the solution reduces to $\phi(x) = 0$ (constant). Include this boundary condition in your program of Exercise 1, and check whether there are more/larger wiggles at a certain grid size, and whether the grid-size limit for central discretisation changes.

The implementation is identical to the previous case, except for the treatment of the outlet boundary condition. Instead of prescribing the value of ϕ at the outlet, a Neumann boundary condition is imposed,

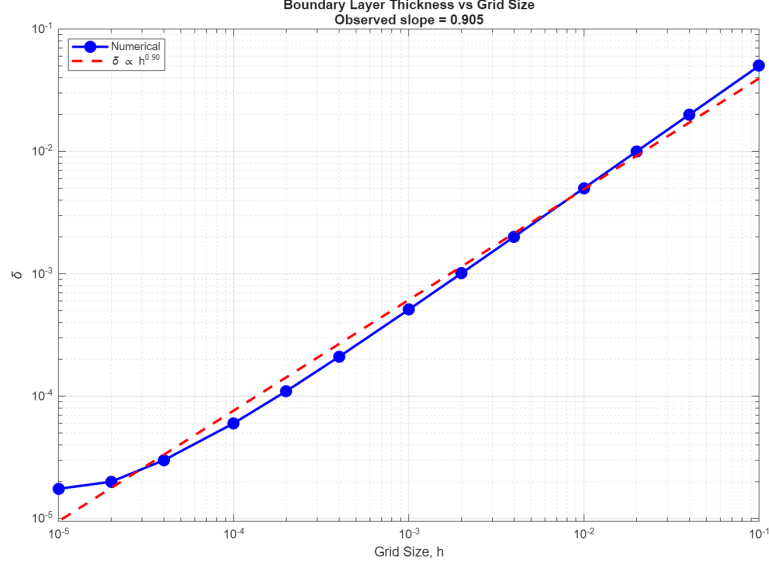


Figure 5: Log-log relation between the measured boundary-layer thickness and the grid spacing for the upwind discretisation.

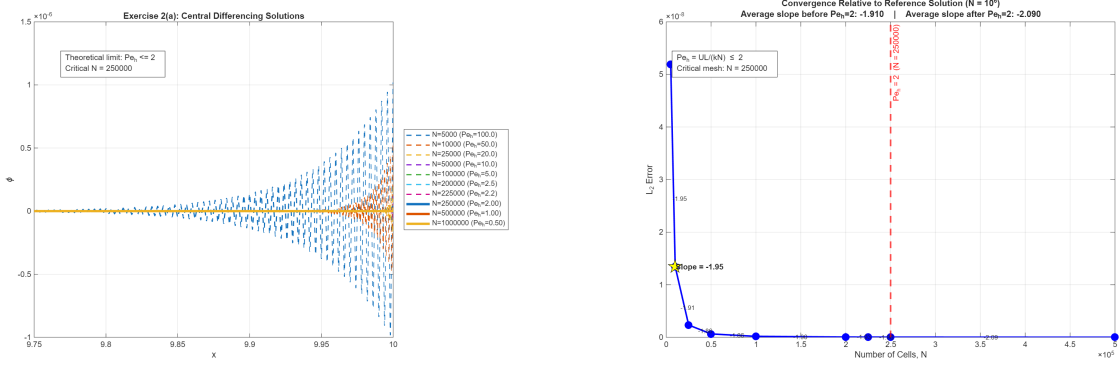


Figure 6: Influence of mesh resolution on central differencing using a Neumann outlet boundary condition. Left: numerical solution profiles near the outlet boundary. Right: convergence plot relative to the reference solution obtained with $N = 10^6$.

$$\left. \frac{\partial \phi}{\partial x} \right|_{x=L} = g.$$

A value of $g = 10^{-3}$ was used in the computations. A homogeneous Neumann condition ($g = 0$) was not considered, as it would lead to the trivial solution for the present problem setup.

The resulting solution profiles and convergence history are shown in Figure 6. As expected, oscillations are still observed for coarse meshes when the cell Péclet number exceeds the monotonicity limit of the central differencing scheme. Therefore, changing the boundary condition does not remove the wiggle phenomenon. The appearance of wiggles is still governed by the sign of the discretisation coefficients and the cell Péclet number criterion

$$Pe_h \leq 2.$$

However, unlike the Dirichlet-Dirichlet case, the convergence plot exhibits a nearly second-order slope from the beginning of the refinement process. This behaviour is not related to the particular value of the Neumann boundary condition but to the accuracy of the discretisation itself.

Central differencing approximates both the first and second spatial derivatives with a truncation error of $O(h^2)$. Furthermore, the Neumann boundary condition is implemented using a consistent discretisation that also introduces an error of order $O(h^2)$. Consequently, the complete discrete problem, including both the interior equations and the boundary treatment, remains second-order accurate. Therefore, once the error is measured with respect to a sufficiently refined reference solution, the dominant error behaves as

$$\|\phi_h - \phi\| \propto h^2.$$

- Now take $\phi'(10) = 20$ as boundary condition. This also gives the system more freedom, but will not give a trivial solution. Again check whether this affects the occurrence of wiggles.

As said in the previous point, and as we can see in Figure 7, we still have wiggles.

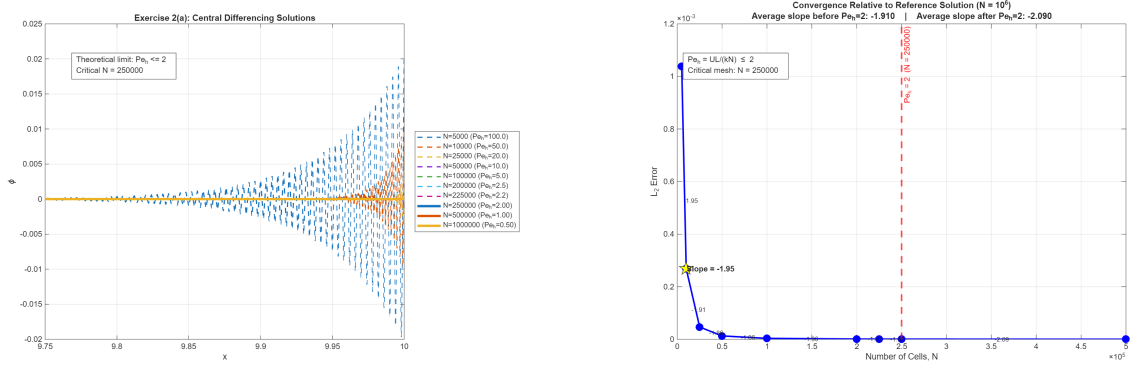


Figure 7: Influence of mesh resolution on central differencing using a Neumann outlet boundary condition. Left: numerical solution profiles near the outlet boundary. Right: convergence plot relative to the reference solution obtained with $N = 10^6$.

- Explain the situation when $\phi'(0) = \phi'(10) = 0$. What happens to the matrix in this case?

Imposing homogeneous Neumann boundary conditions at both ends, means that only the gradient of the solution is prescribed, while no value of ϕ is specified. Consequently, the solution is only determined up to an arbitrary additive constant. Indeed, if $\phi(x)$ is a solution, then $\phi(x) + C$ is also a solution for any constant C , since adding a constant does not change the derivatives. Therefore, the problem does not possess a unique solution. In the discrete system, this lack of uniqueness is reflected in the coefficient matrix. The rows of the matrix become linearly dependent and the matrix loses rank. Equivalently, the matrix possesses a zero eigenvalue and its determinant vanishes. Hence, the matrix is singular and cannot be inverted. Numerical solvers such as TDMA will fail. Looking at the physical aspect a homogeneous Neumann condition at both boundaries implies that no flux enters or leaves the domain.

Exercise 3 – Accuracy Study

Consider the original problem with Dirichlet conditions

$$\phi(0) = \phi_0, \quad \phi(L) = \phi_1. \quad (6)$$

Take “nice” values for L , U , k and ϕ_0, ϕ_1 , such that

$$Pe = 500. \quad (7)$$

- Show by means of a number of numerical simulations (i.e. runs of your program of Exercise 1) that the central method has 2nd order accuracy everywhere, and the (regular) upwind method has 1st order accuracy everywhere. Use the analytic solution as reference solution.
- How can you determine the order of accuracy numerically when the analytic solution is not known? Try this.

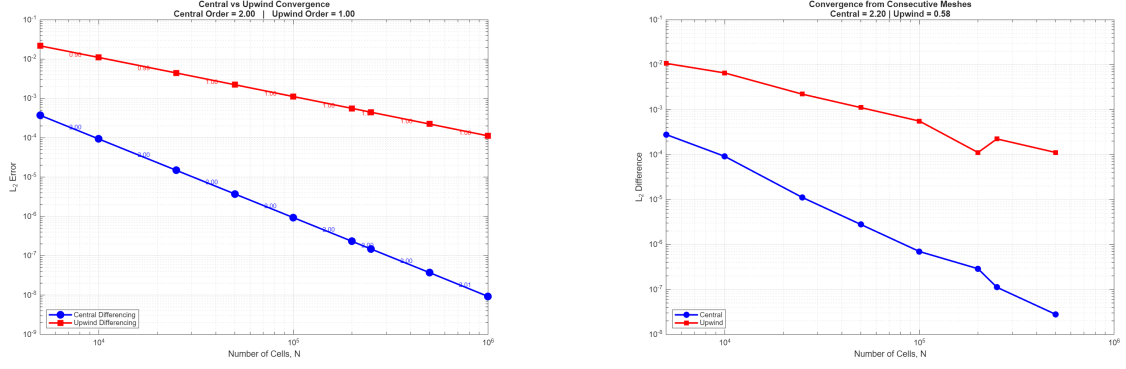


Figure 8: Comparison of the central and upwind discretisation schemes. Left: convergence relative to the analytical solution. Right: self-convergence study based on two consecutive meshes. The central differencing scheme exhibits second-order convergence, whereas the upwind scheme exhibits first-order convergence.

The order of accuracy was determined using two independent approaches. The first approach employs the analytical solution, whereas the second approach uses only numerical solutions obtained on successively refined meshes. When the analytical solution is available, the discretisation error on a mesh with characteristic size h is defined as

$$e_h = \|\phi_h - \phi_{\text{exact}}\|_2.$$

Assuming that the numerical method is of order p , the error behaves according to

$$e_h = Ch^p,$$

where C is a constant independent of the mesh size. For two consecutive meshes h_i and h_{i+1} ,

$$e_i = Ch_i^p,$$

and

$$e_{i+1} = Ch_{i+1}^p.$$

Taking the ratio of the two errors gives

$$\frac{e_{i+1}}{e_i} = \left(\frac{h_{i+1}}{h_i}\right)^p.$$

Taking the natural logarithm yields

$$\ln\left(\frac{e_{i+1}}{e_i}\right) = p \ln\left(\frac{h_{i+1}}{h_i}\right).$$

Therefore, the observed order of accuracy is computed as

$$p = \frac{\ln(e_{i+1}/e_i)}{\ln(h_{i+1}/h_i)}.$$

This expression was applied to the sequence of errors obtained by comparing the numerical solutions with the analytical solution. The central differencing scheme has a truncation error of order $O(h^2)$, and is therefore expected to satisfy

$$p_{\text{central}} = 2.$$

The upwind differencing scheme has a truncation error of order $O(h)$, and is therefore expected to satisfy

$$p_{\text{upwind}} = 1.$$

Agreement between the orders obtained from the analytical-error study and the self-convergence study provides a strong verification of the theoretical accuracy of both discretisation schemes.

3 Exercise 4 – Artificial Diffusion

Files and commands

Required files:

- `cfd_1.m`
- `cfd_upw.m`
- `ns_exact.m`

The program is started in Matlab with the command

`cfd_1`

Description

Consider the inhomogeneous convection-diffusion equation ($U = 1$)

$$\frac{d\phi}{dx} - k \frac{d^2\phi}{dx^2} = S \quad (8)$$

on $0 \leq x \leq 1$, with

$$\phi(0) = 0, \quad \phi(1) = 1. \quad (9)$$

The right-hand side is given by

$$S(x) = \begin{cases} 2 - 5x, & 0 \leq x \leq 0.4, \\ 0, & 0.4 < x \leq 1. \end{cases} \quad (10)$$

In this exercise an upwind discretisation of the above equation is applied. The discrete solution for three values of the diffusion coefficient k is plotted in one picture. Also the exact solution for one value of k can be included in this picture.

Questions

1. Firstly, study the discrete solution for a grid with $N = 20$ segments, at the following values of k :

$$k = 0.1, \quad k = 0.01, \quad k = 0.001. \quad (11)$$

These values correspond with mesh-Péclet numbers

$$P = 0.5, \quad 5, \quad 50, \quad (12)$$

respectively. One would expect (or at least hope) that these solutions tend to the (analytical) solution for $k = 0$. Hence, compare the discrete solutions to the exact solution at $k = 0$ (ignore the Matlab warnings concerning division by zero). What do you observe; is there any such relation?

In Figure 9, the numerical upwind solutions are compared with the analytical solution corresponding to $k = 0$. As expected the results show that there is only partial convergence to the exact solution. The source term forces the solution to increase in the first part of the domain and then remain approximately constant once the source term vanishes. This behaviour is clearly reproduced by all numerical solutions, and as k decreases the profiles indeed move closer to the exact $k = 0$ solution. Nevertheless, the convergence towards the inviscid solution is not complete. Even for $k = 0.001$, the numerical solution does not coincide exactly with the analytical solution. In particular, the plateau value between $x \approx 0.4$ and $x \approx 0.95$ remains below the exact value of 0.4, and the sharp increase near $x = 1$ is considerably smeared. Instead of a nearly discontinuous jump

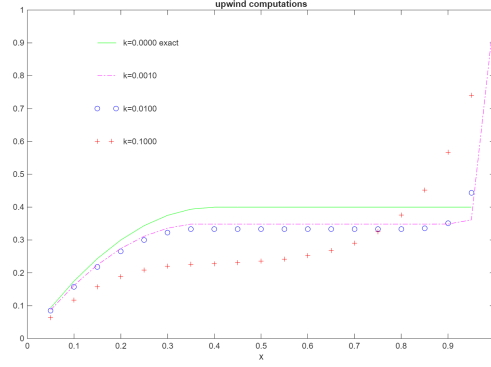


Figure 9: Upwind $k = 0.1, 0.01, 0.001$.

at the outlet, the upwind solutions exhibit a gradual increase spread over several grid cells. This behaviour is a direct consequence of the artificial diffusion introduced by the upwind discretisation. When the physical diffusion coefficient k becomes small, the numerical diffusion associated with the discretisation does not vanish. As a result, the scheme effectively solves a more diffusive problem than the original equation. The influence of this artificial diffusion is thus clearly visible near $x = 1$ and in the plateau, where the exact solution contains a very sharp rise while the numerical solutions smooth this rise over a finite region and the plateau value is below the source term value. Consequently, although the solutions move towards the $k = 0$ profile as k decreases, they do not converge exactly to it on the fixed grid because the numerical diffusion remains present.

2. Next, compare the same three discrete solutions at

$$k = 0.1, \quad k = 0.01, \quad k = 0.001 \quad (13)$$

(again for $N = 20$) to the exact solution at $k = 0.025$. Now, there seems to be some relation between the upwind limit for $k \rightarrow 0$ and the exact solution at $k = 0.025$. Explain this relation.

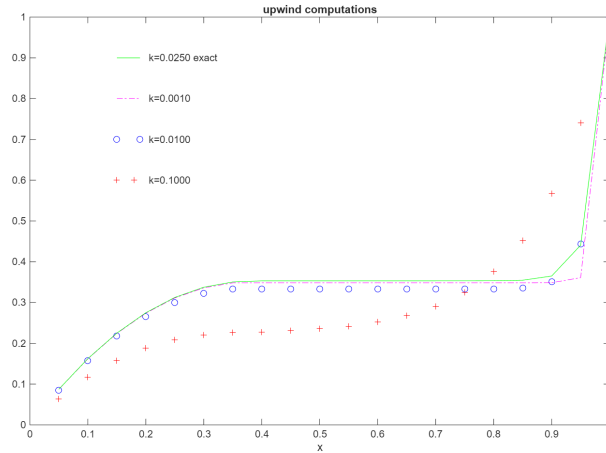


Figure 10: Upwind $k = 0.1, 0.01, 0.001$. Exact $k = 0.025$

Figure 10 compares the same upwind solutions with the analytical solution corresponding to $k = 0.025$. In contrast to the previous comparison with the inviscid solution ($k = 0$), a good agreement is now observed, especially for the case $k = 0.001$. Both the plateau value and the smeared outlet layer are reproduced very accurately by the exact solution with $k = 0.025$. This behaviour can be explained by the artificial diffusion introduced by the upwind discretisation. The first-order upwind scheme does not solve the original differential equation exactly, but rather a modified equation containing an additional diffusion term. The numerical diffusion coefficient is given by

$$k_{\text{num}} = \frac{U \Delta x}{2}.$$

For the present case,

$$U = 1, \quad N = 20, \quad \Delta x = \frac{1}{20} = 0.05,$$

which yields

$$k_{\text{num}} = \frac{1 \cdot 0.05}{2} = 0.025.$$

Therefore, the upwind discretisation effectively solves the problem with an increased diffusion coefficient

$$k_{\text{eff}} = k + k_{\text{num}}.$$

As $k \rightarrow 0$,

$$k_{\text{eff}} \rightarrow 0.025.$$

Consequently, the limiting solution of the upwind scheme is not the exact solution with $k = 0$, but rather the exact solution corresponding to

$$k = 0.025.$$

This is precisely what is observed in Figure 10. The solution for $k = 0.001$ almost coincides with the analytical solution for $k = 0.025$, while the solutions for $k = 0.01$ and $k = 0.1$ remain more diffused because the physical diffusion is still significant compared to the numerical diffusion.

3. Finally, study once more the upwind solutions at

$$k = 0.1, \quad k = 0.01, \quad k = 0.001, \quad (14)$$

but now on a grid with $N = 40$ segments. Which exact solution, i.e. at which value of k , corresponds with the upwind limit for $k \rightarrow 0$?

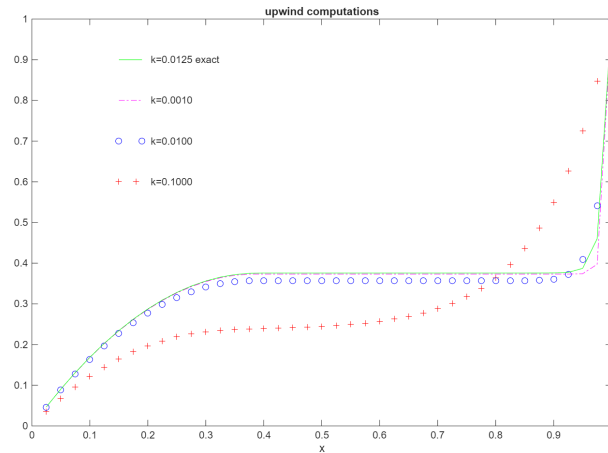


Figure 11: Upwind $k = 0.1, 0.01, 0.001$. Exact $k = 0.0125$

Figure 11 shows the numerical upwind solutions obtained on the refined grid with $N = 40$ cells together with the analytical solution corresponding to $k = 0.0125$. The relation between these two results can be understood from the artificial diffusion introduced by the upwind discretisation. As

shows before, the upwind scheme therefore behaves as if it were solving a problem with an effective diffusion coefficient

$$k_{\text{eff}} = k + k_{\text{num}},$$

where

$$k_{\text{num}} = \frac{1 \times 0.025}{2} = 0.0125.$$

As the physical diffusion coefficient tends to zero,

$$k \rightarrow 0,$$

the effective diffusion tends to

$$k_{\text{eff}} \rightarrow 0.0125.$$

Consequently, the limiting upwind solution is expected to coincide with the exact solution corresponding to

$$\boxed{k = 0.0125}.$$

This prediction is confirmed by Figure 11. The solution computed with $k = 0.001$ lies very close to the analytical solution for $k = 0.0125$, reproducing both the plateau region generated by the source term and the sharp increase near the outlet boundary. Compared with the case $N = 20$, the amount of artificial diffusion has been reduced by a factor of two, from 0.025 to 0.0125. As a result, the numerical solutions move closer to the inviscid limit and the outlet layer becomes sharper.